

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10.

1 Write 0.00000350793 in scientific notation correct to three significant figures.

- (A) 0.351×10^{-5}
- (B) 3.508×10^{-6}
- (C) 35.1×10^{-7}
- (D) 3.51×10^{-6}

2 Solve $|2x + 1| \leq 5$.

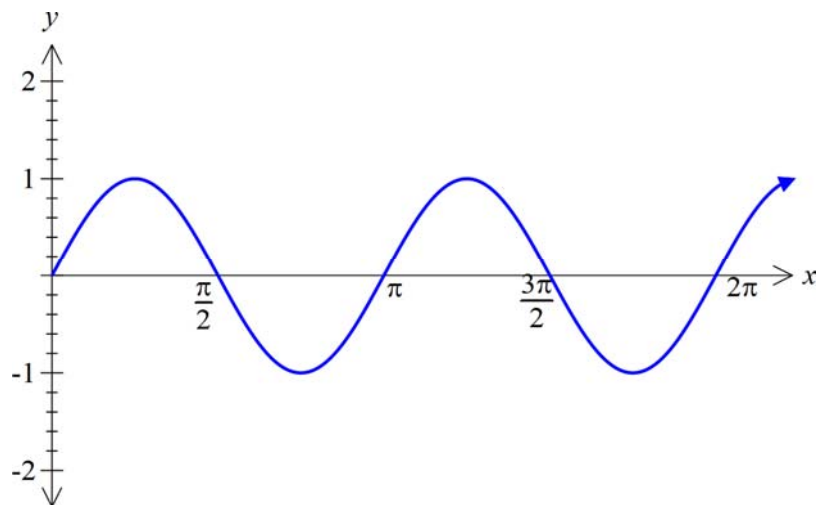
- (A) $-3 \leq x \leq 2$
- (B) $-2 \leq x \leq 3$
- (C) $x \geq 2, x \leq -3$
- (D) $x \geq 3, x \leq -2$

3 A parabola has equation $(y - 3) = 8(x + 2)^2$. The coordinates of its vertex are

- (A) $(3, -2)$
- (B) $(-3, 2)$
- (C) $(-2, 3)$
- (D) $(2, -3)$

Multiple Choice (continued).

- 4 Choose the equation of the graph below.



- (A) $y = 2 + \sin x$
(B) $y = 2 \sin x$
(C) $y = \sin 2x$
(D) $y = \sin(x + 2)$

- 5 Find $\frac{1}{2} \int e^{\frac{t}{2}} dt$.

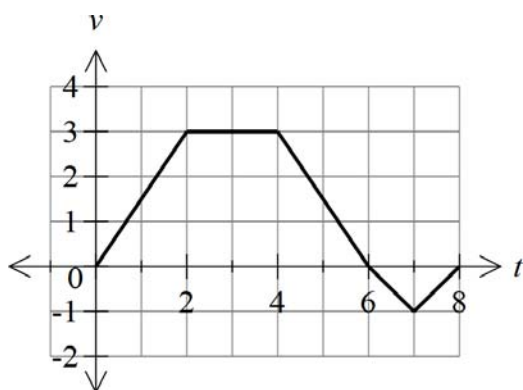
- (A) $e^{-\frac{t}{2}} + C$
(B) $e^{\frac{t}{2}} + C$
(C) $2e^{\frac{t}{2}} + C$
(D) $e^t + C$

Multiple Choice (continued).

6 If $\int_a^b f(x) dx = a + 2b$, then $\int_a^b (f(x) + 5) dx$ is given by

- (A) $a + 2b + 5$
- (B) $a + 2b - 5$
- (C) $7b - 5a$
- (D) $7b - 4a$

Questions 7 and 8 refer to the following graph:



A bug begins to crawl up a vertical wire at time $t = 0$. The velocity v cm/s of the bug at time t seconds, $0 \leq t \leq 8$, is given by the function whose graph is shown above.

7 After how many seconds does the bug change direction?

- (A) 2
- (B) 4
- (C) 6
- (D) 7

8 What is the total distance the bug travelled from $t = 0$ to $t = 8$?

- (A) 13 cm
- (B) 11 cm
- (C) 8 cm
- (D) 4 cm

Multiple Choice (continued).

- 9 The quadratic function $y = 2x^2 - 5x + 1$ has
- (A) No real roots.
 - (B) Two real, distinct, rational roots.
 - (C) Two real, distinct, irrational roots.
 - (D) Two equal roots.
-
- 10 If $e^y = \sin x$, $0 \leq x \leq \pi$, find $\frac{dy}{dx}$.
- (A) $\tan x$
 - (B) $\cot x$
 - (C) $-\cot x$
 - (D) $-\tan x$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a **separate** writing booklet.

(a) Find a and b such that $\frac{1}{2\sqrt{3}-1} = a\sqrt{3} + b$. 2

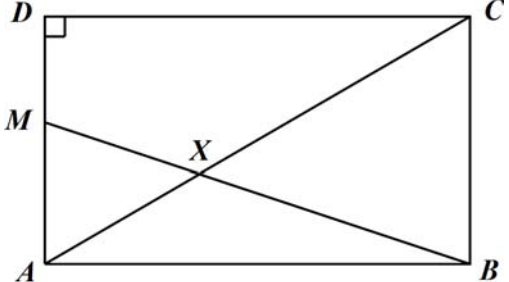
(b) Solve $\frac{x-1}{2} - \frac{2x-3}{3} < 1$. 2

(c) Find the equation, in general form, of the normal to the curve $y = \frac{1-x}{1+x}$ at the point where $x = 0$. 3

(d) Find a primitive function of $\frac{x^2}{x^3-2}$. 2

(e) Find the exact value of x for which $5^{x+2} = 7$. 2

(f) The area of a sector of a circle is 28.86 cm^2 and contains an angle at the centre of $\frac{\pi}{6}$ radians. 2
Find the radius of the circle, correct to one decimal place.

(g)  $ABCD$ is a rectangle. 2
The line BM meets AC at X .
Prove that the triangles AXM and CXB are similar.

Question 12 (15 marks) Use a **separate** writing booklet.

(a) Differentiate with respect to x .

(i) $\tan x$ **1**

(ii) $\frac{\ln x}{x}$ **1**

(iii) $\frac{1}{7x+4}$ **2**

(b) Find $\int \sqrt{5x-2} \, dx$. **2**

(c) Zoe is saving for a holiday. In the first month she saves \$20 and in the second month she saves \$25. In each subsequent month her savings are \$5 more than the month before.

(i) How much will she save in the 18th month? **2**

(ii) How much money will she have saved in total by the end of the 18th month? **1**

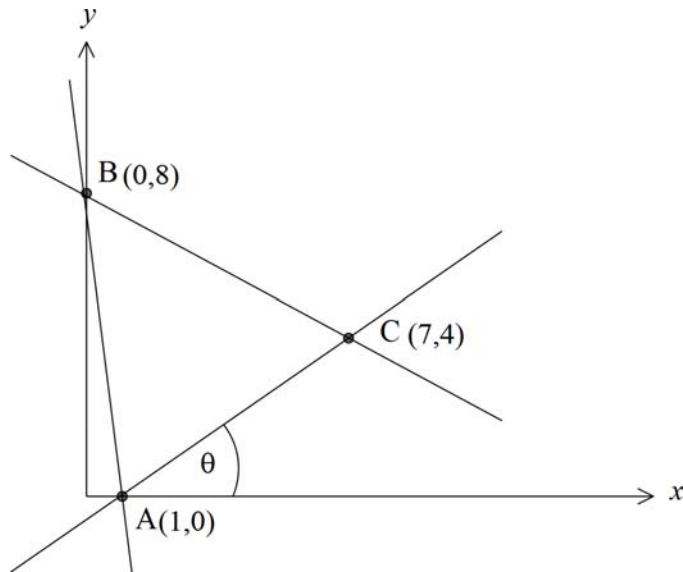
(iii) Zoe needs \$1900 to pay for her plane ticket. How many whole months will it take for her to save at least this amount? **3**

(d) Use Simpson's rule with 5 function values to find an approximation to **3**

$\int_1^{17} \sqrt{x} \, dx$, giving your answer correct to 2 decimal places.

Question 13 (15 marks) Use a **separate** writing booklet.

(a)



The points A , B and C have coordinates $(1,0)$, $(0,8)$ and $(7,4)$ respectively.
The angle between the line AC and the positive direction of the x -axis is θ .

- | | | |
|-------|---|---|
| (i) | Find the gradient of the line AC . | 1 |
| (ii) | Calculate the size of angle θ , correct to the nearest minute. | 1 |
| (iii) | Find the coordinates of D , the midpoint of AC . | 1 |
| (iv) | Show that AC is perpendicular to BD . | 1 |
| (v) | Find the area of $\triangle ABC$. | 3 |

(b) Consider the curve $y = x^3 - 6x^2 + 9x$.

- | | | |
|-------|---|---|
| (i) | Find the coordinates of the stationary points and determine their nature. | 4 |
| (ii) | Find the coordinates of the point(s) of inflexion. | 2 |
| (iii) | Draw a neat sketch of the curve, showing all important features. | 2 |

Question 14 (15 marks) Use a **separate** writing booklet.

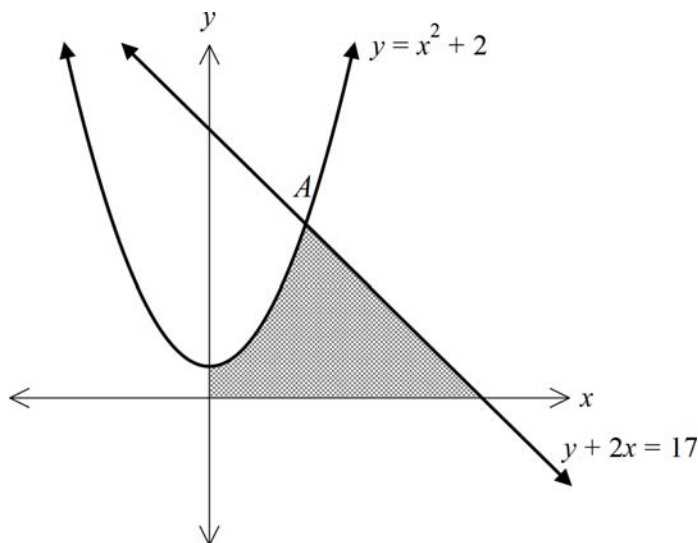
(a) Evaluate $\int_0^{\frac{3\pi}{2}} \cos 2\theta \, d\theta$. 2

- (b) Sophie planted a seed in the ground. During the first year it grew 50 cm. Each year after that the plant grew $\frac{2}{3}$ of the previous year's growth. 2

What is the maximum height of the plant?

(c) Solve $\sin 2x + 1 = 0$ for $0 \leq x \leq 2\pi$. 2

- (d) The figure below shows part of the curve $y = x^2 + 2$ and the line $y + 2x = 17$.



- (i) Show that the coordinates of A are $(3, 11)$. 1

- (ii) Find the area of the shaded region. 3

Question 14 continues on next page.

Question 14 (continued).

- (e) The rate of decay of a radioactive substance is proportional to the mass present at time t years and

is given by $\frac{dM}{dt} = -kM$.

- (i) Show that $M = M_0 e^{-kt}$ satisfies the equation $\frac{dM}{dt} = -kM$. **1**

- (ii) If it takes 24 000 years for half of the substance to decay, show that $k = \frac{\ln 2}{24\,000}$. **2**

- (iii) How long would it take for $\frac{2}{3}$ of this substance to decay? Give your answer correct to the nearest thousand years. **2**

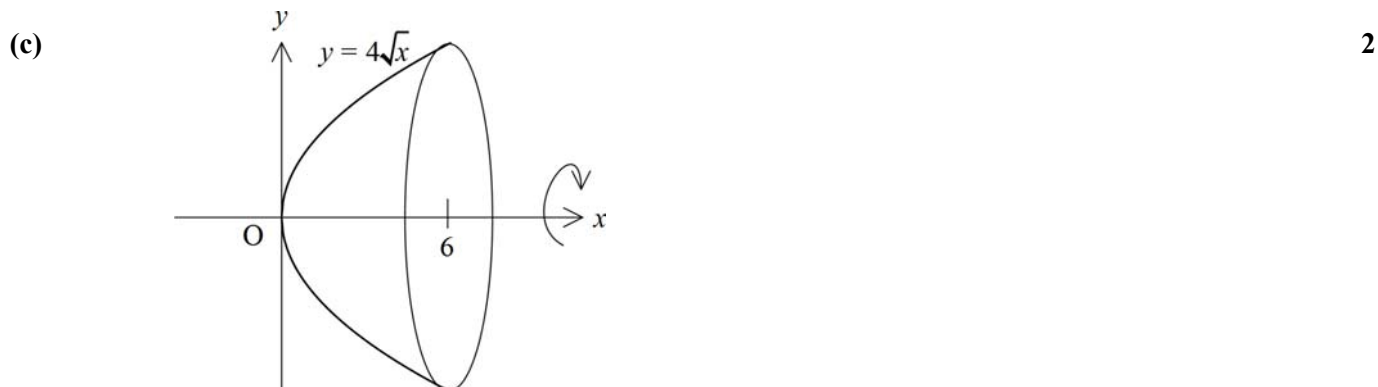
Question 15 (15 marks) Use a **separate** writing booklet.

- (a) Given that $\frac{d^2y}{dx^2} = 2 - \frac{4}{x^2}$ and $\frac{dy}{dx} = 3$ at the point (1,6), find y in terms of x . 3

- (b) A particle moving in a straight line on the x -axis has displacement x metres after t seconds given by the function

$$x = 2t^2 - 19t + 35.$$

- (i) What is the initial position of the particle? 1
- (ii) What is the initial velocity of the particle? 1
- (iii) At what times was the particle at the origin? 2
- (iv) At what time was the particle instantaneously at rest? 1
- (v) How far did the particle travel in between its visits to the origin? 2



The region enclosed by the curve $y = 4\sqrt{x}$ and the x axis between $x = 0$ and $x = 6$ is rotated about the x axis as shown.

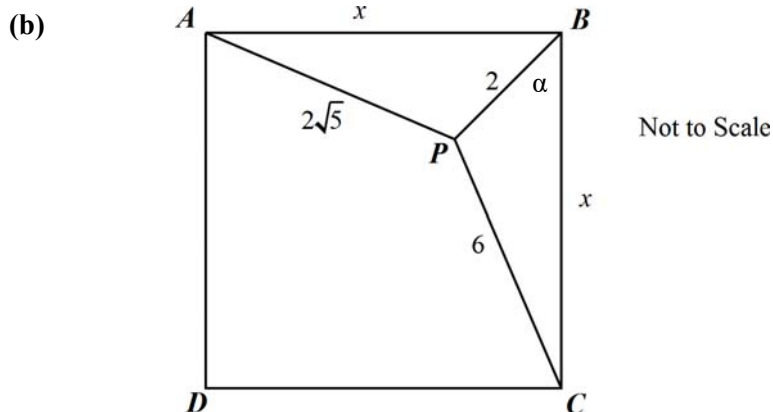
Find the volume of the solid of rotation.

- (d) Prove $(\cot \theta + \operatorname{cosec} \theta)^2 = \frac{1 + \cos \theta}{1 - \cos \theta}$. 3

Question 16 (15 marks) Use a **separate** writing booklet.

- (a) The current of I amperes flowing in an electrical circuit is given by $I = \frac{dQ}{dt}$, where Q is the charge on a capacitor and t is the time in seconds. For a particular electrical circuit, $Q = 4 \sin \pi t$. 2

Calculate the current flowing in the circuit after 2 seconds.



The diagram shows a square $ABCD$ of side length x cm. P is a point inside the square, such that $PC = 6$ cm, $PB = 2$ cm and $AP = 2\sqrt{5}$ cm.

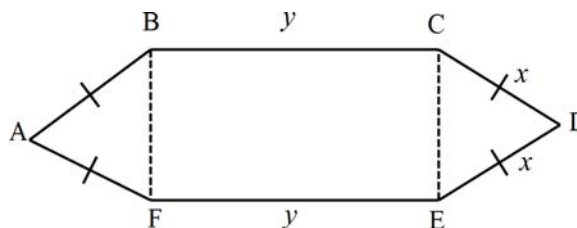
Let $\angle PBC = \alpha$.

- (i) Using the cosine rule in triangle PBC , show that $\cos \alpha = \frac{x^2 - 32}{4x}$. 1
- (ii) By considering triangle PBA , show that $\sin \alpha = \frac{x^2 - 16}{4x}$. 2
- (iii) Hence, or otherwise, show $x^4 - 56x^2 + 640 = 0$. 2
- (iv) Find the only solution for x , giving reasons. 2

Question 16 continues on next page.

Question 16 (continued).

- (c) A wire of length 120 cm is bent to form the perimeter of a frame $ABCDEF$, where ABF and CDE are equilateral triangles of side x cm and $BC = EF = y$ cm.



- | | | |
|-------|---|----------|
| (i) | Express y in terms of x . | 1 |
| (ii) | Show that the total area of $ABCDEF$, $Z \text{ cm}^2$, is given by $Z = \left(\frac{\sqrt{3}}{2} - 2 \right) x^2 + 60x$. | 2 |
| (iii) | Find the value of x for which Z will be a maximum, leaving your answer in the form $a + b\sqrt{3}$, where a and b are constants. | 3 |

End of paper

2013 Mathematics Trial Solutions.

1. $3.50793 \times 10^{-6} = 3.51 \times 10^{-6}$ D

2. $|2x+1| \leq 5$
 $-5 \leq 2x+1 \leq 5$
 $-6 \leq 2x \leq 4$
 $-3 \leq x \leq 2$ A.

3. $(y-3) = 8(x+2)^2$
 $(-2, 3)$ C

4. $y = \sin 2x$ C

5. $\frac{1}{2} \int e^{t/2} dt$
 $= e^{t/2} + c$ B

6. $\int_a^b f(x) dx = a + 2b$

$\int_a^b (f(x) + 5) dx$
 $= [5x]_a^b + a + 2b$
 $= 5b - 5a + a + 2b$
 $= 7b - 4a$ D.

7. 6 sec C.

8. 13 cm. A

9. $y = 2x^2 - 5x + 1$
 $\Delta = (-5)^2 - 4 \times 2 \times 1$
 $= 17$ $\therefore$ Two real, distinct irrational C

10. $e^y = \sin x$ $\frac{dy}{dx} = \frac{\cos x}{\sin x} = \cot x$ B
 $y = \ln(\sin x)$

Question 11

$$a) \frac{1}{2\sqrt{3}-1} = a\sqrt{3} + b.$$

$$\frac{1}{2\sqrt{3}-1} = \frac{1}{2\sqrt{3}-1} \times \frac{2\sqrt{3}+1}{2\sqrt{3}+1}$$
$$= \frac{2\sqrt{3}+1}{12-1} \quad (1)$$

$$= \frac{2\sqrt{3}}{11} + \frac{1}{11}$$

$$a = \frac{2}{11} \quad b = \frac{1}{11} \quad (1)$$

$$b) \frac{x-1}{2} - \frac{2x-3}{3} < 1$$

$$\frac{3(x-1) - 2(2x-3)}{6} < 1$$

$$3x - 3 - 4x + 6 < 6 \quad (1)$$

$$-x + 3 < 6$$

$$-x < 3$$

$$x > -3 \quad (1)$$

$$c) \quad y = \frac{1-x}{1+x} \quad u = 1-x \quad v = 1+x$$

$$u' = -1 \quad v' = 1$$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$= \frac{(1+x)(-1) - (1-x)(1)}{(1+x)^2}$$

$$= \frac{-1-x-1+x}{(1+x)^2}$$

$$= \frac{-2}{(1+x)^2} \quad (1)$$

When $x = 0$

$$\frac{dy}{dx} = \frac{-2}{(1+0)^2}$$
$$= -2$$

ie gradient of tangent is $m_T = -2$

$$\therefore \text{gradient of normal } m_N = -\frac{1}{m_T}$$
$$= \frac{1}{2} \quad (1)$$

When $x = 0$

$$y = \frac{1-0}{1+0}$$

$$= 1 \quad \text{ie } (0, 1)$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{2}(x - 0)$$

$$2y - 2 = x$$

$$x - 2y + 2 = 0 \quad (1)$$

d) $\int \frac{x^2}{x^3 - 2} dx$

$$= \frac{1}{3} \int \frac{3x^2}{x^3 - 2} dx$$

$$= \frac{1}{3} \ln(x^3 - 2) + C$$

① ①

e) $5^{x+2} = 7$

$$(x+2) \ln 5 = \ln 7$$

$$x+2 = \frac{\ln 7}{\ln 5}$$

$$x = \frac{\ln 7}{\ln 5} - 2$$

OR $5^{x+2} = 7$

$$x+2 = \log_5 7$$

$$x = \log_5 7 - 2$$

(2)

f) $A = 28.86$ $\theta = \frac{\pi}{6}$

Area of sector $A = \frac{1}{2} r^2 \theta$

$$28.86 = \frac{1}{2} \times r^2 \times \frac{\pi}{6} \quad (1)$$

$$r^2 = \frac{28.86 \times 12}{\pi}$$

$$= 110.2370798$$

$$r = \sqrt{110.23707 \dots}$$

$$= 10.49938473$$

$$\approx 10.5 \text{ cm.} \quad (1)$$

(correct rounding required)

g) $AM \parallel CB$ (opposite sides of a rectangle are parallel) (1)

$\angle MAX = \angle BCX$ (alternate angles in parallel lines are equal; $AM \parallel CB$)

$\angle AXM = \angle CXB$ (vertically opposite angles are equal) (1)

$\therefore \triangle AXM \parallel \triangle CXB$ (equiangular)

Question 12.

a) (i) $\frac{d}{dx} \tan x = \sec^2 x$ ①

(ii) $\frac{d}{dx} \left(\frac{\ln x}{x} \right)$

$u = \ln x \quad v = x$

$u' = \frac{1}{x} \quad v' = 1$

$= \frac{vu' - uv'}{v^2}$

$= \frac{x \times \frac{1}{x} - \ln x \times 1}{x^2}$

$= \frac{1 - \ln x}{x^2}$ ①

(iii) $\frac{d}{dx} \left(\frac{1}{7x+4} \right)$

$= \frac{d}{dx} (7x+4)^{-1}$

$= -(7x+4)^{-2} \times 7$

$= \frac{-7}{(7x+4)^2}$ ①

b) $\int \sqrt{5x-2} \, dx$

$= \int (5x-2)^{1/2} \, dx$

$= \frac{2}{3} \times \frac{(5x-2)^{3/2}}{5}$ ① + C

$= \frac{2(5x-2)^{3/2}}{15} + C$

$= \frac{2}{15} \sqrt{(5x-2)^3} + C$

$$c) \quad T_1 = \$20 \quad a = 20$$

$$T_2 = \$25 \quad d = 5$$

$$T_3 = \$30 \quad \text{is arithmetic sequence} \quad (1)$$

$$(i) \quad T_n = a + (n-1)d$$

$$T_{18} = 20 + (18-1) \times 5$$

$$= \$105 \quad (1)$$

$$(ii) \quad S_n = \frac{n}{2}(a+L) \quad L = 105$$

$$S_{18} = \frac{18}{2}(20+105)$$

$$= \$1125 \quad (1)$$

$$(iii) \quad S_n \geq \$1900$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$\frac{n}{2}(2 \times 20 + (n-1) \times 5) \geq 1900 \quad (1)$$

$$n(40 + 5n - 5) \geq 3800$$

$$35n + 5n^2 \geq 3800$$

$$n^2 + 7n \geq 760$$

$$n^2 + 7n - 760 \geq 0$$

$$\text{Solve } n^2 + 7n - 760 = 0$$

$$n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-7 \pm \sqrt{7^2 - 4 \times 1 \times 760}}{2 \times 1}$$

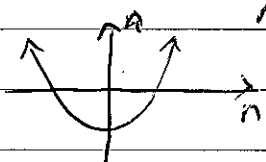
$$= \frac{-7 \pm \sqrt{3089}}{2}$$

$$n = -31.289 \dots \quad \text{or } n = 24.289 \dots \quad (1)$$

ie since n is a positive whole number,

$$n \geq 25$$

$\therefore$ 25 whole months. (1)



d) $f(x) = \sqrt{x}$ $\int_1^{17} \sqrt{x} \, dx$

x	1	5	9	13	17
$\sqrt{x}$	1	$\sqrt{5}$	3	$\sqrt{13}$	$\sqrt{17}$

 $h = \frac{17-1}{4} = 4$

$$\int_1^{17} \sqrt{x} \, dx \approx \frac{4}{3} \left((1 + \sqrt{17}) + 2 \times 3 + 4 \times (\sqrt{5} + \sqrt{13}) \right) \textcircled{1}$$

$$= 45.986 \dots$$

$$\approx 45.99 \textcircled{1}$$





AC is perpendicular
to BP

$$(v) \quad A = \frac{1}{2}bh$$

base is length AC
height is length BD.

$$\begin{aligned} AC &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(7-1)^2 + (4-0)^2} \\ &= \sqrt{6^2 + 4^2} \\ &= \sqrt{52} \quad (1) \end{aligned}$$

$$\begin{aligned} BD &= \sqrt{(4-0)^2 + (2-8)^2} \\ &= \sqrt{4^2 + (-6)^2} \\ &= \sqrt{52} \quad (1) \end{aligned}$$

$$A = \frac{1}{2}bh$$

$$\begin{aligned} &= \frac{1}{2} \times \sqrt{52} \times \sqrt{52} \\ &= 26 \text{ units}^2 \quad (1) \end{aligned}$$

$$b) \quad y = x^3 - 6x^2 + 9x.$$

$$(1) \quad \frac{dy}{dx} = 3x^2 - 12x + 9.$$

Stationary points when $\frac{dy}{dx} = 0$.

$$3x^2 - 12x + 9 = 0 \quad (1)$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 3 \text{ or } x = 1 \quad (1)$$

$$\begin{aligned} \text{When } x = 3, \quad y &= 3^3 - 6 \times 3^2 + 9 \times 3 \\ &= 27 - 54 + 27 \\ &= 0 \end{aligned}$$

ie (3, 0) } (1)

$$\begin{aligned} \text{When } x = 1, \quad y &= 1 - 6 + 9 \\ &= 4 \end{aligned}$$

ie (1, 4)

$$\frac{d^2y}{dx^2} = 6x - 12$$

$$\text{At } (3, 0) \quad \frac{d^2y}{dx^2} = 6 \times 3 - 12$$

$$\frac{d^2y}{dx^2} = 6 > 0$$

ie concave up

$(3, 0)$ is a minimum turning point.

$$\text{At } (1, 4) \quad \frac{d^2y}{dx^2} = 6 \times 1 - 12$$

$$\frac{d^2y}{dx^2} = -6 < 0$$

ie concave down.

$\therefore (1, 4)$ is a maximum turning point.

(ii) For points of inflexion

$$\frac{d^2y}{dx^2} = 0$$

$$6x - 12 = 0$$

$$x = 2.$$

$$\text{When } x = 2, \quad y = 2^3 - 6 \times 2^2 + 9 \times 2$$

$$= 8 - 24 + 18$$

$$= 2.$$

ie $(2, 2)$ (1)

Check concavity

x	1.5	2	2.5
$\frac{d^2y}{dx^2}$	-3	0	3

$\therefore$ Since concavity changes at $(2, 2)$ (1)
it is a point of inflexion

(iii) x -intercepts when $y=0$

$$0 = x^3 - 6x^2 + 9x$$

$$= x(x^2 - 6x + 9)$$

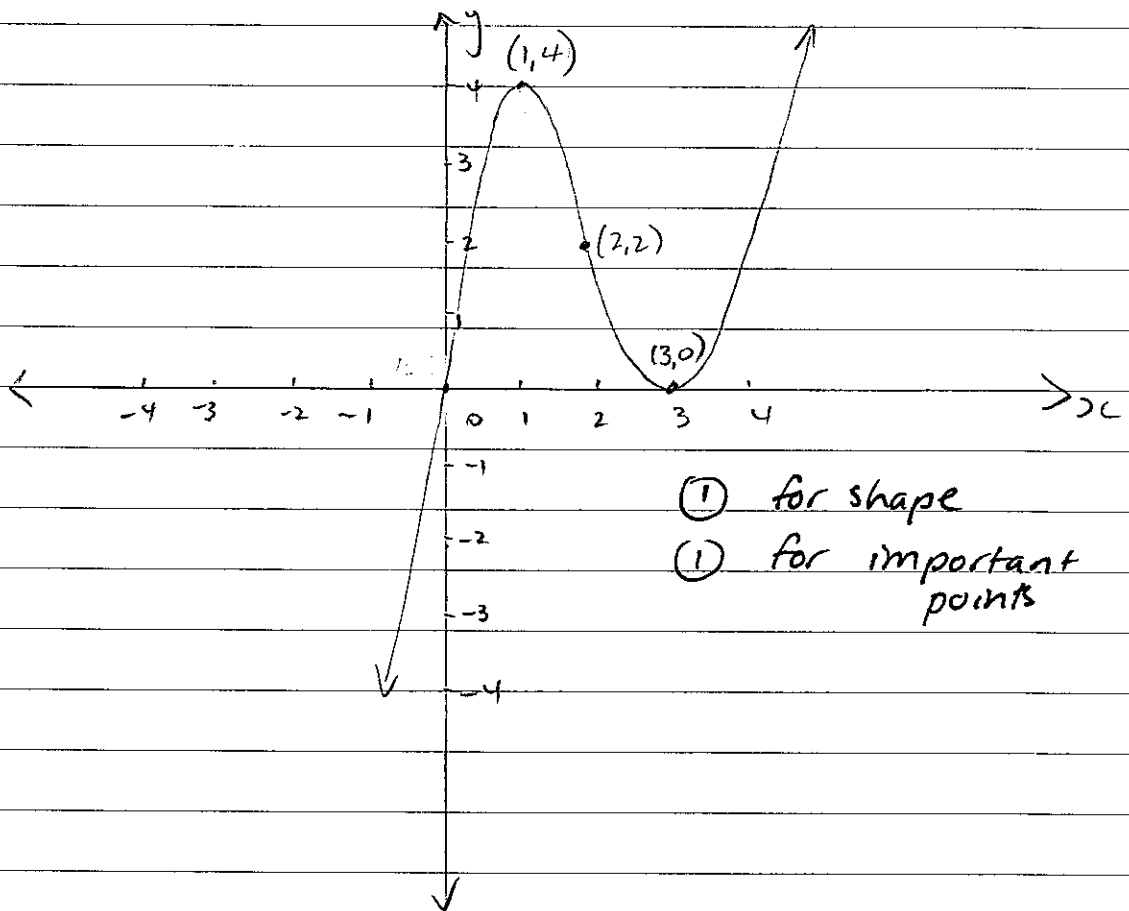
$$= x(x-3)^2$$

$$x=0 \quad \text{or} \quad x=3$$

i.e. x -intercepts at 0 or 3.

y -intercepts when $x=0$

i.e. at 0.



Question 14.

$$a) \int_0^{\frac{3\pi}{2}} \cos 2\theta \, d\theta$$
$$= \left[\frac{1}{2} \sin 2\theta \right]_0^{\frac{3\pi}{2}} \quad (1)$$

$$= \frac{1}{2} \left(\sin 2 \times \frac{3\pi}{2} - \sin 2 \times 0 \right)$$

$$= \frac{1}{2} (\sin 3\pi - \sin 0)$$

$$= 0 \quad (1)$$

b) $a = 50$ $r = \frac{2}{3}$

This is geometric series with $|r| < 1$

$$S = \frac{a}{1-r}$$
$$= \frac{50}{1-\frac{2}{3}} \quad (1)$$
$$= \frac{50}{\frac{1}{3}}$$
$$= 150 \quad (1)$$

$\therefore$ maximum height of plant is 150cm.

c) $\sin 2x + 1 = 0$

$$\sin 2x = -1$$

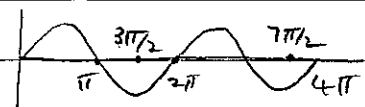
$$0 \leq x \leq 2\pi$$

Let $\theta = 2x$

$$0 \leq \theta \leq 4\pi$$

$$\sin \theta = -1$$

$$\therefore \theta = \frac{3\pi}{2}, \frac{7\pi}{2}$$



$$\therefore x = \frac{3\pi}{4} \quad \text{or} \quad \frac{7\pi}{4} \quad (2)$$

d) (1) Solve simultaneously

$$\begin{array}{rcl} y + 2x & = & 17 \quad \text{---(1)} \\ y & = & x^2 + 2 \quad \text{---(2)} \end{array}$$

Substitute (2) into (1)

$$x^2 + 2 + 2x = 17.$$

$$x^2 + 2x - 15 = 0.$$

$$(x - 3)(x + 5) = 0$$

$$x = 3 \quad \text{or} \quad x = -5$$

Substitute $x = 3$ into (2)

$$\begin{aligned} y &= 3^2 + 2 \\ &= 11 \end{aligned}$$

Since A is in first quadrant,
(3, 11) is the point we require.

(ii) Find where $y + 2x = 17$ crosses x-axis.
 $0 + 2x = 17$
 $x = 8.5$

$$\text{Area} = \int_0^3 (x^2 + 2) dx + \int_3^{8.5} (-2x + 17) dx \quad \text{①}$$

$$= \left[\frac{x^3}{3} + 2x \right]_0^3 + \left[-\frac{2x^2}{2} + 17x \right]_3^{8.5} \quad \text{①}$$

$$= \left(\frac{3^3}{3} + 2 \times 3 \right) - (0 + 0) + \left(-8.5^2 + 17 \times 8.5 \right) - \left(-3^2 + 17 \times 3 \right)$$

$$= (9 + 6) + 72.25 - 42$$

$$= 45.25 \text{ units}^2 \quad \text{①}$$

$$e) \quad \frac{dM}{dt} = -kM$$

$$(i) \quad M = M_0 e^{-kt}$$

$$\begin{aligned} \frac{dM}{dt} &= M_0 \times -k e^{-kt} \\ &= -k M_0 e^{-kt} \quad (1) \\ &= -k M \end{aligned}$$

$$(ii) \quad \text{When } t = 24000 \quad M = \frac{M_0}{2}$$

$$\text{ie } \frac{M_0}{2} = M_0 e^{-k \times 24000}$$

$$\frac{1}{2} = e^{-k \times 24000} \quad (1) \quad \text{OR}$$

$$2 = e^{k \times 24000}$$

$$\ln \frac{1}{2} = -k \times 24000$$

$$k \times 24000 = \ln 2$$

$$\ln 2^{-1} = -k \times 24000$$

$$k = \frac{\ln 2}{24000}$$

$$- \ln 2 = -k \times 24000$$

$$k = \frac{\ln 2}{24000}$$

$$(iii) \quad M = \frac{M_0}{3} \quad \text{find } t = ?$$

$$\frac{1}{3} M_0 = M_0 e^{-kt}$$

$$\frac{1}{3} = e^{-kt} \quad (1)$$

$$3 = e^{kt}$$

$$\ln(3) = kt$$

$$k = \frac{\ln 2}{24000}$$

$$t = \frac{\ln(3)}{\frac{\ln 2}{24000}}$$

$$= \frac{\ln 2}{24000}$$

$$= 38039.10002$$

$$\text{ie } 38000 \text{ years (nearest thousand)} \quad (1)$$

Question 15.

$$a) \frac{d^2y}{dx^2} = 2 - \frac{4}{x^2}$$

$$\frac{dy}{dx} = 3 \quad \text{at} \quad (1, 6)$$

$$\frac{d^2y}{dx^2} = 2 - 4x^{-2}$$

$$\frac{dy}{dx} = 2x - \frac{4x^{-1}}{-1} + C$$

$$= 2x + \frac{4}{x} + C.$$

$$\text{At } (1, 6) \quad \frac{dy}{dx} = 3$$

$$\therefore 3 = 2 \times 1 + \frac{4}{1} + C.$$

$$\underline{C = -3.}$$

$$\therefore \frac{dy}{dx} = 2x + \frac{4}{x} - 3$$

$$= 2x + 4x^{-1} - 3 \quad (1)$$

$$y = x^2 + 4 \ln x - 3x + C.$$

$$(1, 6) \quad 6 = 1^2 + 4 \ln 1 - 3 \times 1 + C. \quad (1)$$

$$6 = 1 + 0 - 3 + C$$

$$6 = -2 + C$$

$$\underline{C = 8}$$

$$\therefore y = x^2 + 4 \ln x - 3x + 8. \quad (1)$$

$$6) \quad x = 2t^2 - 19t + 35$$

(i) when $t=0$, $x=35$
 $\therefore$ initial position is 35 m. (1)

(ii) $\frac{dx}{dt} = 4t - 19$

when $t=0$, $\frac{dx}{dt} = -19$

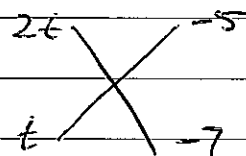
$\therefore$ initial velocity is -19 m/s (1)

(iii) $x=0$

$$2t^2 - 19t + 35 = 0$$

$$(2t-5)(t-7) = 0 \quad (1)$$

$$t = 2.5 \text{ or } t = 7$$



$\therefore$ particle is at origin
 when $t = 2.5$ seconds and
 $t = 7$ seconds. (1)

(iv) instantaneously at rest when
 $\frac{dx}{dt} = 0$

$$4t - 19 = 0$$

$$4t = 19$$

$$t = 19/4$$

$$= 4\frac{3}{4} \text{ seconds. (1)}$$

(v) At $t = 4\frac{3}{4}$

$$x = 2 \times \left(\frac{19}{4}\right)^2 - 19 \times \left(\frac{19}{4}\right) + 35$$

$$= -10.125 \quad (1)$$

$\therefore$ distance travelled $= 2 \times 10.125$
 $= 20.25 \text{ m. (1)}$

$$e) y = 4\sqrt{x}$$

$$V = \pi \int_0^6 y^2 dx$$

$$= \pi \int_0^6 (4\sqrt{x})^2 dx \quad (1)$$

$$= \pi \int_0^6 16x dx$$

$$= 16\pi \left[\frac{x^2}{2} \right]_0^6$$

$$= 8\pi (6^2 - 0^2)$$

$$= 8\pi \times 36$$

$$= 288\pi \text{ units}^3 \quad (1)$$

$$d) (\cot\theta + \operatorname{cosec}\theta)^2 = \frac{1 + \cos\theta}{1 - \cos\theta}$$

$$\text{LHS} = (\cot\theta + \operatorname{cosec}\theta)^2$$

$$= \cot^2\theta + 2\cot\theta \operatorname{cosec}\theta + \operatorname{cosec}^2\theta$$

$$= \frac{\cos^2\theta}{\sin^2\theta} + 2 \frac{\cos\theta}{\sin\theta} \times \frac{1}{\sin\theta} + \frac{1}{\sin^2\theta} \quad (1)$$

$$= \frac{\cos^2\theta + 2\cos\theta + 1}{\sin^2\theta}$$

$$= \frac{(\cos\theta + 1)^2}{1 - \cos^2\theta} \quad (1)$$

$$= \frac{(1 + \cos\theta)^2}{(1 - \cos\theta)(1 + \cos\theta)} \quad (1)$$

$$= \frac{1 + \cos\theta}{1 - \cos\theta}$$

$$= \text{RHS}$$

Question 16.

a) $Q = 4 \sin \pi t$

$$I = \frac{dQ}{dt} = 4 \times \pi \cos \pi t$$
$$= 4\pi \cos \pi t \quad (1)$$

When $t = 2$ $I = 4\pi \cos(\pi \times 2)$

$$= 4\pi \times 1$$
$$= 4\pi \text{ amperes.} \quad (1)$$

b) (i) In $\triangle PBC$

$$\cos \alpha = \frac{PC^2 + 2^2 - 6^2}{2 \times 2 \times PC} \quad (1)$$
$$= \frac{x^2 + 4 - 36}{4x}$$
$$= \frac{x^2 - 32}{4x}$$

(ii) In $\triangle PBA$ $\angle ABP = 90^\circ - \alpha$.

$$\cos(90^\circ - \alpha) = \frac{x^2 + 2^2 - (2\sqrt{5})^2}{2 \times 2 \times x}$$
$$= \frac{x^2 + 4 - 4 \times 5}{4x}$$
$$= \frac{x^2 - 16}{4x} \quad (1)$$

Also, $\cos(90^\circ - \alpha) = \sin \alpha \quad (1)$

$$\therefore \sin \alpha = \frac{x^2 - 16}{4x}$$

$$(iii) \sin^2 x + \cos^2 x = 1$$

$$\left(\frac{x^2-16}{4x}\right)^2 + \left(\frac{x^2-32}{4x}\right)^2 = 1 \quad (1)$$

$$\frac{(x^4 - 32x^2 + 256) + (x^4 - 64x^2 + 1024)}{16x^2} = 1$$

$$2x^4 - 96x^2 + 1280 = 16x^2 \quad (1)$$

$$\div 2 \quad 2x^4 - 112x^2 + 1280 = 0$$

$$x^4 - 56x^2 + 640 = 0$$

$$(iv) \text{ let } u = x^2$$

$$u^2 - 56u + 640 = 0$$

$$(u - 40)(u - 16) = 0$$

$$u = 40 \quad \text{or} \quad u = 16$$

$$x^2 = 40 \quad \text{or} \quad x^2 = 16$$

$$x = \pm\sqrt{40} \quad x = \pm 4 \quad (1)$$

$$= \pm 2\sqrt{10}$$

x must be positive since it is a length.

The solution must be $2\sqrt{10}$ cm since for any triangle with sides a, b and c
 $c < a + b$.

$$\text{If } x = 4$$

$$6 < 2 + 4 \quad (\text{false})$$

$$\text{If } x = 2\sqrt{10}$$

$$6 < 2 + 2\sqrt{10} \quad (\text{true})$$

$$2 < 6 + 2\sqrt{10} \quad (\text{true})$$

$$2\sqrt{10} < 6 + 2 \quad (\text{true})$$

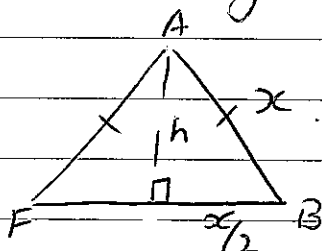
(1)

$$c) \quad (1) \quad 2y + 4x = 120$$

$$y + 2x = 60$$

$$y = 60 - 2x \quad (1)$$

(ii)



Using Pythagoras'

$$x^2 = h^2 + (x/2)^2$$

$$x^2 = h^2 + x^2/4$$

$$h^2 = x^2 - x^2/4$$

$$h^2 = \frac{3x^2}{4}$$

$$h = \frac{\sqrt{3}x}{2} \quad (1)$$

$$\therefore Z = xy + 2x + \frac{1}{2}x \times \frac{\sqrt{3}}{2}x$$

$$= x(60 - 2x) + \frac{\sqrt{3}}{2}x^2 \quad (1)$$

$$= 60x - 2x^2 + \frac{\sqrt{3}}{2}x^2$$

$$= \left(\frac{\sqrt{3}}{2} - 2\right)x^2 + 60x$$

$$(iii) \quad \frac{dZ}{dx} = \left(\frac{\sqrt{3}}{2} - 2\right) \times 2x + 60 \quad (1)$$

For maximum $\frac{dZ}{dx} = 0$

$$2\left(\frac{\sqrt{3}}{2} - 2\right)x + 60 = 0$$

$$2\left(\frac{\sqrt{3}}{2} - 2\right)x = -60$$

$$\left(2 - \frac{\sqrt{3}}{2}\right)x = 30$$

$$\left(\frac{4 - \sqrt{3}}{2}\right)x = 30$$

$$\begin{aligned}
 x &= \frac{30 \times 2}{(4 - \sqrt{3})} \times \frac{4 + \sqrt{3}}{4 + \sqrt{3}} \\
 &= \frac{240 + 60\sqrt{3}}{16 - 3} \\
 &= \frac{240 + 60\sqrt{3}}{13} \quad (1) \\
 &= \frac{240}{13} + \frac{60\sqrt{3}}{13}
 \end{aligned}$$

$$\begin{aligned}
 \frac{d^2Z}{dx^2} &= 2\left(\frac{\sqrt{3}}{2} - 2\right) \\
 &= -2.2679 \dots
 \end{aligned}$$

Since $\frac{d^2Z}{dx^2} < 0$, concave down (1)

So maximum Z for x value given above.

OR Using quadratic method.
 $Z = \left(\frac{\sqrt{3}}{2} - 2\right)x^2 + 60x$

$$\begin{aligned}
 \text{Axis of symmetry } x &= -\frac{b}{2a} \\
 &= \frac{-60}{2\left(\frac{\sqrt{3}}{2} - 2\right)} \\
 &= \frac{60}{4 - \sqrt{3}} \times \frac{4 + \sqrt{3}}{4 + \sqrt{3}} \\
 &= \frac{240 + 60\sqrt{3}}{16 - 3} \quad (1) \\
 x &= \frac{240}{13} + \frac{60\sqrt{3}}{13}
 \end{aligned}$$

Z is a maximum here since the coefficient of x^2 is $\frac{\sqrt{3}}{2} - 2 = -1.1339 \dots < 0$

$\therefore$ (parabola is concave down)